Human senses

If a person is to survive and function properly, he or she must respond to changes in the environment. Anything that causes a response from us is called a *stimulus* (plural *stimuli*). We detect stimuli from both inside and outside our bodies and act upon them. The parts of the body that detect these stimuli are called *receptors*.

1 Name three types of stimuli that a person might experience in normal living.

Receptors are groups of specialised cells that respond to stimuli. Sense organs, eg eyes, ears, tongue, nose and skin, contain many receptors. A stimulus causes these cells to send information to other parts of the body using a *messenger*, either through the bloodstream or by electrical pulses in the nervous system. When the signal reaches an *effector* at a specific location, there is a response. The effector, usually a muscle or a gland, does something.

2 Describe how a muscle might respond when it receives a signal.

3 Describe how a gland might respond when it receives a signal.

The sequence is: stimulus → receptor → messenger → effector → response.

The table below shows some stimuli, how they are detected and the sensation or response they cause.

Stimulus	Location of receptor	Sensation/response
light	eye	sight
sound	ear	hearing
warmth	skin	warmth
cold	skin	cold
touch	skin	pressure/pain
chemicals	mouth, nose	taste, smell

Copy and complete the table. The first one is done for you.

Stimulus	Sense organ	Response
speech	ear	turn towards speaker
bright light	4	blink
heat	5	sweat
6	mouth	taste
chemicals	7	smell
8	skin	pressure
gravity	9	balance

10 Identify the definition that best describes a **stimulus.**

A An incentive for a person to do something.
B Something that excites a sensory organ, causes a muscle to contract or a gland to secrete.
C A chemical messenger sent by a gland through the bloodstream.
D Part of the body that registers changes in the immediate environment.

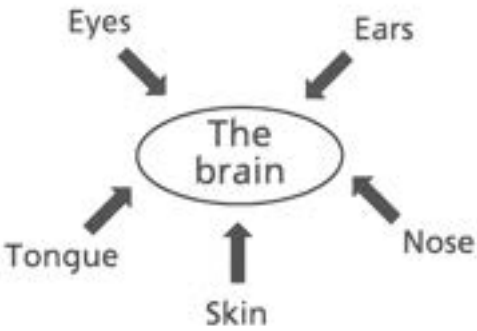
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Research

11 Describe what *sensory deprivation* is and how it affects humans.

12 The colours of our surroundings are supposed to influence our moods. Outline some theories about this and give some examples.



The brain is being bombarded by messages from the various sensory systems all of the time. It has to sort out which messages it can ignore from those which are important. Then it has to decide what to do about the important ones. It could store the information or it could choose between several different ways of reacting. It is this ability to think and make complicated choices that makes our brain different from that of other animals.

Sensory receptors are made up of several groups. Copy the following table and state what the receptors sense and where they are located.

Receptor type	What receptor responds to	Location of receptor
chemoreceptor	**13**	**14**
hair cells	**15**	**16**
mechanoreceptor	**17**	**18**
photoreceptor	**19**	**20**
thermoreceptor	**21**	**22**

We use our senses to take in information from our external environment. One way we do this is by communicating with each other. Sound is used when we talk to each other.

23 Give a definition for *communicate*.

24 Give two other examples where (non-speech) sound is used by humans to communicate.

It is likely that when you describe your surroundings, the sense that you use the most is *sight*. It has been estimated that in humans, of the information that the brain uses, 70% comes from sight, 10% from smell, 5% from taste and 5% from touch. Sight is used in communication when we read or look at pictures or gestures etc.

Visible light is sometimes referred to as 'white light'. It is the part of the electromagnetic spectrum that the human eye uses to see things. The eye collects visible light, changes this to electrical signals and these are carried by nerves to the brain.

In humans, the sense of *smell* is relatively underdeveloped when compared to other animals, for example a dog or a pig, which depend on it to a far greater extent. Substances that have an odour give off molecules, which are present in air and are breathed in. Mucous membrane, a damp lining in the nose, can dissolve these small particles. The special nerve endings present pass the message to the brain.

State whether the following statements are true or false:

25 'Aural' relates to the ear or to the sense of hearing.

26 Our ears contain balance receptors.

27 The most important human sense is touch.

28 You have been watching a programme on TV about 'ESP – Extra Sensory Perception'. In this show some people claimed they could read other people's minds. You would like to test to see how likely this is. Describe an experiment you could carry out, using, for example, shapes on a series of cards.

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